

A resampling method to search for gravitational waves from light primordial black holes binaries

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I scienced something, so now you get to read it.

I. INTRODUCTION

Primordial black holes (PBHs) are a type of black hole proposed to have formed during the early universe. They differ from astrophysical black holes which form from stellar collapse. [NL: Recent observations of black holes provide tantalizing hints that PBHs may exist.](#) LVK citations There has been the observation of four BBH signals with subsolar components although none achieve a significance sufficient to claim detection [? ?]. Additionally, a significant portion of black holes observed with gravitational waves have low effective spins [? ?] which is not expected for astrophysical black holes which tend to form with higher spins [?] and suggests that a portion of these black holes could be primordial [?]. Black holes have also been observed with masses within the pair-instability gap ($60M_{\odot} - 150M_{\odot}$) [? ? ? ?] and lower-mass gap ($2.5M_{\odot} - 5M_{\odot}$) [? ? ? ?] which are mass regimes that should not be populated by first generation astrophysical black holes [? ? ?]. In addition, the mass distribution of observed black holes has an excess around $35M_{\odot}$ [? ?], which is not well understood and could suggest separate astrophysical and primordial black hole populations [?]. See [?] for a review of other possible observational evidence of PBHs.

These observational hints have revitalised theoretical interest in PBHs which were previously proposed [?] but considered unlikely to exist [? ? ?]. Recent analyses suggest the existence of PBHs could explain the characteristics of black holes observed with gravitational waves and also help solve many of the open problems in astronomy and cosmology. PBHs are a form of dark matter because they are formed before the big-bang nucleosynthesis (BBN) and are not included in the matter estimates derived from the BBN [? ?] so could form a fraction, or all, of the dark matter in the universe [? ?]. PBHs could have acted as the seeds for supermassive black holes which exist in the centres of galaxies and whose formation mechanism is not well understood [? ? ?]. They could also be sensitive to the physics of the early universe [?] and probe the high-energy present then that is otherwise inaccessible [?].

The formation mechanism(s) and initial mass distribution of primordial black holes is not known. Any process producing sufficiently dense regions in the early universe would have lead to gravitational collapse and be a creation mechanism for PBHs. Proposed formation mechanisms for

PBHs include; collapses from primordial inhomogeneities in the initial mass-energy distribution of the universe [? ?], perturbations in the potential of the inflationary field [? ?], or a QCD phase transition which abruptly modifies the critical density due to a global change of pressure [? ?], see [? ?] for reviews of PBH formation mechanisms. Depending on their formation mechanism, primordial black holes could have formed with masses between $[10^{-18} - 10^9]M_{\odot}$ [?]. No direct observations of PBHs currently exist, so searches for observational signatures need to cover a wide mass range. Observational limits on primordial black holes constrain their abundance for different possible masses but often assume a monochromatic mass function, which is not guaranteed in many formation scenarios [? ?].

After formation, primordial black holes could form binaries. This could have happened in the early universe if two PBHs were sufficiently close together so that they decouple from the Hubble flow and became gravitationally bound [? ? ?]. PBHs could also form binaries in the later universe through dynamical capture [? ? ?], three-body interactions [?], or a 'catalysis' mechanism [?], but such mechanisms are likely subdominant compared to the early-universe formation channel. See [?] for a recent review.

After PBH binaries form, they would emit gravitational waves and their evolution would be similar to astrophysical black holes where they inspiral, merge, and form a remnant black hole which emits a post-merger ringdown signal. However, determining the origins of stellar mass black holes ($M \gtrsim 5M_{\odot}$) is difficult because observational signatures of astrophysical versus primordial production are not well understood, although see [1]. Observation of a subsolar mass black hole, however, would be considered definitive evidence of its primordial origins as standard stellar mechanisms cannot produce black holes with such low masses. Driven by the detection of solar-mass compact binary coalescence (CBC) events, multiple searches for sub-solar mass events with masses between $[0.2, 1]M_{\odot}$ have been performed by both the LVK [2–5] and independently [6–8] which find no or weak evidence for detections. These searches are performed with matched filter based pipelines. These approaches, however, have issues scaling to lower masses because signal durations become longer, increasing the computational cost of matched filtering [9].

Less attention has been dedicated to gravitational wave signals from lighter PBH binaries which emit gravita-

tional waves with frequencies within the sensitivity band of LVK detectors during their inspiral phase. Only a small number of search have been performed including a search for signals from PBHs with masses (1eX - 1eY) in [10], upper limits imposed in [11], and new techniques recently developed in [12? , 13]. These methods are all semi-coherent, where computational cost is decreased by dividing the data into several chunks and incoherently combining detection statistics measured from each segment.

In this paper we demonstrate a new fully coherent method of searching for gravitational waves from light primordial black hole inspirals using resampling.

This paper is organised as follow ...

[Micro-lensing events somewhere?](#)

II. SIGNAL MODEL

The frequency evolution of gravitational waves emitted by a PBH binary during their inspiral can be calculated in the same way as for conventional black hole binaries; by equating the gravitational wave power to the loss of orbital energy, to derive:

$$f(\beta, t) = f_0 \left(1 - \frac{8}{3}\beta t\right)^{-3/8}, \quad (1)$$

where f_0 is the frequency at the reference time $t = 0$, and we have introduced the parameter:

$$\beta = \frac{96}{5}\pi^{8/3} \left(\frac{G}{c^3}\right)^{5/3} M_c^{5/3} f_0^{8/3}, \quad (2)$$

where $M_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$ is the chirp mass of the binary system and m_1 and m_2 are the masses of each black hole. The corresponding phase evolution of this signal is:

$$\phi(\beta, t) = -\frac{6\pi}{5} f_0 \frac{(1 - \frac{8}{3}\beta t)^{5/8}}{\beta} + \phi_0. \quad (3)$$

The time required for the system to evolve from an initial frequency f_0 to a final frequency f_1 can be derived by rearranging Eq. 1 to get:

$$t_{0 \rightarrow 1} = \frac{0.375}{\beta} \left(1 - \left(\frac{f_0}{f_1}\right)^{8/3}\right). \quad (4)$$

The amplitude evolution of the binary system is:

$$h_0(t) = \frac{4}{d} \left(\frac{GM_c}{c^2}\right)^{5/3} \left(\frac{\pi f(t)}{c}\right)^{2/3} \quad (5)$$

$$\simeq 1.6 \times 10^{-24} \left(\frac{d}{10\text{kpc}}\right)^{-1} \left(\frac{M_c}{10^{-3}M_\odot}\right)^{5/3} \left(\frac{f}{50\text{Hz}}\right)^{2/3} \quad (6)$$

The inspiral phase of a binary black hole system lasts until it reaches the frequency of the inner-most circular orbit (ISCO), which can be approximated as:

$$f_{\text{ISCO}} \simeq 2200 \left(\frac{M_\odot}{M_{\text{tot}}}\right) \text{Hz}, \quad (7)$$

For light primordial black holes with chirp masses below a solar mass, the ISCO frequency is above the sensitivity band of current audio-band detectors (like the LVK detectors) so any detected signal would be of the inspiral. Note that Eqs. 1 - 7 use the quadrupole approximation which is valid during the inspiral when the black holes are far from coalescence [?]. This signal model also assumes that the PBH binary is not precessing or eccentric which should be the case for all PBHs formed in the early universe [1, 14]

In addition to the frequency and amplitude *evolution* of the waveform due to the inspiral, there is also frequency and amplitude *modulation* of the signal when observed by gravitational wave detectors on Earth. The frequency modulation is due to the Doppler effect caused by the relative motion of the Earth and the PBH binary. The amplitude modulation is caused by the time-varying detector antennae patterns. [Should the analytic amplitude and frequency modulation be given here or discussed in the corresponding sections?](#)

III. THE RESSAMPLING METHOD

In this section we describe the basic concept of resampling and discuss an improved implementation of it using non-uniform fast Fourier transforms. Resampling is a data analysis technique used to transform a signal with known frequency evolution into a monochromatic signal. It was originally proposed for gravitational wave signals in [15] and specifically for continuous gravitational waves in [16–18] and an implementation is described in [19]. It is used for several continuous wave pipelines used by the LVK collaboration (citation?).

The basic idea of resampling is to define a new time variable which the signal is monochromatic with respect to so that the signal can efficiently be analysed using Fast-Fourier Transforms (FFTs). For PBH binaries, we define the new time variable:

$$\tau(\beta) = -\frac{3}{5} \frac{(1 - \frac{8}{3}\beta t)^{5/8}}{\beta}, \quad (8)$$

substituting Eq. 8 into Eq. 3 gives:

$$\phi(\beta, \tau) = 2\pi f_0 \tau(\beta), \quad (9)$$

i.e. the signal is monochromatic with respect to τ . A demonstration of the basic principle of resampling is shown in figure ???. The location of the peak in the power spectrum identifies the f_0 of the signal can be combined with knowledge of the β which correctly resamples the signal to calculate M_c .

The signal is not uniformly sampled with respect to the new time variable (τ) which is necessary to be able to compute the standard FFT. Previous implementations of resampling used "stroboscopic" resampling [17, 19?] where a higher sampling frequency is used for the original time series then nearest-neighbour interpolation is used so that the resampled time series is uniformly resampled. Here we instead introduce a new resampling implementation which relies on non-uniform fast Fourier transforms (NUFFT).

NUFFT algorithms are a family of techniques which approximate the Fourier Transform of non-uniformly sampled data to a desired accuracy in numerically efficient ways. Their implementation combines the resampling of the data onto a uniform grid, the implementation of a conventional FFT and some correction factors to achieve the desired accuracy. Many different NUFFT methods and implementations exist which primarily differ based on the technique used to approximation technique used to resample the data which depends on a choice of kernel function. See older review articles with accessible introductions of the subject see [20, 21]. For modern implementations on CPUs and GPUs see [22–26] and the references within. We use the NUFFT algorithm described in [27, 28] and implemented in [22].

To verify the correctness of resampling using a NUFFT, we inject a PBH signal with parameters $M_c = XXM_\odot$, $f_0 = YY$ Hz. We compare the spectra

- Correctness. Power difference 0.3%
- Why quicker
 - Strobo uses $f_{\text{Signal}} = 40 * f_{\text{Max}}$. $f_{\text{Ratio}} = 25$ (check Lorenzo settings).
 - RIT uses a Dirchlet kernel of order 8. Higher order (6 vs 8), each order is more expensive (exp/sqrt vs sin), still needs oversampling
- How much quicker
- Optimal settings / how to choose? - Appendix
- Generalised, can be used for any technique

This allows us to directly compute the Fourier spectrum from the resampled time series which can use a lower sampling rate than that necessary for stroboscopic resampling.

Mention that behaviour of NUFFT will be described in more detail in follow up paper?

Now include the frequency modulation from Doppler

A. Connection to matched filtering

This method is fully coherent and when using the 5-vector statistic is equivalent to matched filtering. If the NUFFT is "provably close to optimal" then this means

that the NUFFT resampling implementation is the optimal tradeoff between computational power and sensitivity.

Need to clarify where do the gains come from, i.e. why is this better than existing matched filter implementations. Probably because evaluating this kernel is easier than computing FFT required for the matched filtering, how does this connect to resampling

B. Connection to heterodyne

Heterodyne relies on the phase, which requires integrating the frequency model. Integration cannot be parallelised while NUFFT can be.

IV. THE 5-VECTOR DETECTION STATISTIC

After resampling, the frequency evolution and modulation from, respectively, the inspiral and Doppler effect will have been removed and the signal will be monochromatic. However, the signal will be amplitude modulated by the time-varying detector antennae patterns. We use the 5-vector detection statistic [29] to account for this amplitude modulation as is common in several CW searches [?]. Here we briefly review the 5-vector statistic and provide clarifications about how it should be applied to binary PBH systems.

A monochromatic signal (e.g. after Doppler corrections and resampling) with constant amplitude will produce a signal at the detector given by:

$$h(t) = h_0(A_+(t)H_+ + A_\times(t)H_\times)e^{i\omega_0 t + \phi_0}, \quad (10)$$

where h_0 is the signal amplitude, ω_0 is the frequency of the monochromatic signal, ϕ_0 is the frequency at the reference time t_0 , $A_{+/\times}(t)$ are periodic functions that vary throughout a sidereal day given in Eqs. (10) - (12) of [?], and:

$$H_+ = \frac{\cos 2\psi - i\eta \sin 2\psi}{\sqrt{1 + \eta^2}}, \quad (11)$$

$$H_\times = \frac{\sin 2\psi + i\eta \cos 2\psi}{\sqrt{1 + \eta^2}}, \quad (12)$$

$$\eta = \frac{-2 \cos \iota}{1 + \cos^2 \iota} \quad (13)$$

where ι is the angle between the plane of the binary system and the line of sight (This is correct for binaries right?) and ψ is the gravitational wave polarisation angle.

The effect of the sidereal modulation is to distribute the signal power into 5 different frequencies: $\omega_0 \pm \mathbf{k}\Omega$, $\mathbf{k} = 0, \pm 1, \pm 2$ where Ω is the sidereal angular frequency of the the Earth. If a gravitational wave detector collects data $x(t)$ over a duration T_{obs} which has a signal with frequency ω_0 present, the data five-vector is given by

(expressed, for simplicity, using the continuous Fourier transform):

$$\mathbf{X} = \int_{T_{\text{obs}}} x(t) e^{-i(\mathbf{k}\Omega + \omega_0)t} dt. \quad (14)$$

We also calculate two template 5-vectors $\mathbf{A}_+$ and $\mathbf{A}_\times$ using Eqs. (17) - (18) of [?]. We can now compute the matched filter for each polarization component:

$$\hat{H}_+ = \frac{\mathbf{X} \cdot \mathbf{A}_+}{|\mathbf{A}_+|^2} \quad (15)$$

$$\hat{H}_\times = \frac{\mathbf{X} \cdot \mathbf{A}_\times}{|\mathbf{A}_\times|^2} \quad (16)$$

which we use for the 5-vector detection statistic:

$$\mathcal{S} = |\mathbf{A}_+|^4 |\hat{H}_+|^2 + |\mathbf{A}_\times|^4 |\hat{H}_\times|^2 \quad (17)$$

This detection statistic optimises the detection probability while minimising the false alarm probability (via a ROC curve) for signals with unknown polarisation parameters (η and ψ). It is far more computationally efficient than computing the matched filter statistic for each possible polarisation configuration which is necessary for CW and even transient CW searches such as for primordial black holes. [Take the N candidates with greatest detection statistic?](#) For the signal(s) with the greatest detection statistic, their amplitude(s) can be estimated by:

$$\hat{h}_0 = \sqrt{|\hat{H}_+|^2 + |\hat{H}_\times|^2} \quad (18)$$

V. OPTIMAL DISTANCE SENSITIVITY

We estimate the maximum distance sensitivity that our method is sensitive to determine whether searches performed on current generation gravitational-wave detector data could plausibly detect primordial black holes. A particular target of interest is the galactic center of the milky way because it hosts a large mass of dark matter [?] which could be used as a proxy for the primordial black holes mass distribution [?], although see [?].

We analytically compute the distance sensitivity by following the general procedure outlined in [?]. For a fixed false-alarm probability (FAP) the threshold critical ratio (CR^*) is:

$$\text{CR}^* = -1 - \ln(\text{FAP}), \quad (19)$$

For a false alarm probability of 10^{-6} , the corresponding critical ratio is $\text{CR}^* = 12.8$. We define the spectral amplitude to be:

$$\lambda = \frac{4|\tilde{h}(f)|^2}{TS_n(f)}, \quad (20)$$

where $\tilde{h}(f)$ is the Fourier transform of the data, $S_n(f)$ is the noise spectral density, and T is the signal duration.

As following Eq. X in ?? we compute the probability of a signal with a given CR and λ :

$$p(\text{CR}, \lambda) = e^{-\text{CR} - 1 - \lambda/2} I_0(\sqrt{2\lambda(\text{CR} + 1)}). \quad (21)$$

The detection probability is now defined to be:

$$P_{\text{det}}(\lambda) = \int_{\text{CR}^*}^{\infty} e^{-\text{CR} - 1 - \lambda/2} I_0(\sqrt{2\lambda(\text{CR} + 1)}) d\text{CR}, \quad (22)$$

where I_0 is the modified Bessel function of the first kind. In practice we fix a detection probability and numerically invert the equation to compute the threshold spectral amplitude λ^* . For a detection probability of 95% (and still using $\text{FAP} = 10^{-6}$), the critical spectral amplitude is $\lambda^* = 46.5$.

The maximum distance sensitivity is then computed by fixing the sky location (here taken to be the galactic centre), averaging over gravitational wave polarisation angle, inclination angle, power loss due to the Fourier window, and time varying detector beam patterns, to get: [probably too much info here](#)

$$d_{\text{opt}} = \frac{0.00757c\beta}{f_0^2 \sqrt{\lambda^*}} \sqrt{\int_{t_0}^{t_0+T} \frac{(1 - \frac{8}{3}\beta t)^{-1/2}}{S_n(f_0(1 - \frac{8}{3}\beta t)^{-3/8})} dt}. \quad (23)$$

We can numerically solve Eq. 23 given a particular noise spectral density. Figure 1 shows the computed distance sensitivity for the O4 design noise spectral density as a function of the signals parameters. The distance sensitivity strongly depends on the signals parameters, varying from $\mathcal{O}(1\text{pc})$ - $\mathcal{O}(10\text{Mpc})$. The red dotted line corresponds to the distance of the galactic centre (8kpc). This demonstrates that for the majority of the parameter space considered, the resampling method has a distance sensitivity that reaches the galactic center. Note that this described the *optimal* distance sensitivity, for any real search where the signal's parameters are not known a-priori, additional signal power will be lost due to mismatches between the true signal parameters and the closest template, this mismatch power loss will be discussed in the following section.

VI. THE SEARCH PARAMETER SPACE

Resampling depends on β and on the start time of the signal??

A. Grid in β

The resampling algorithm is performed using a given value of β (Eq. 8), but its true value is not known a-priori and needs to be searched over. The range of β which needs to be searched over corresponds to the range of signal frequencies f_0 and chirp masses M_c that we aim to search for. For signals with frequencies between $f_0 = [20, 200]\text{Hz}$

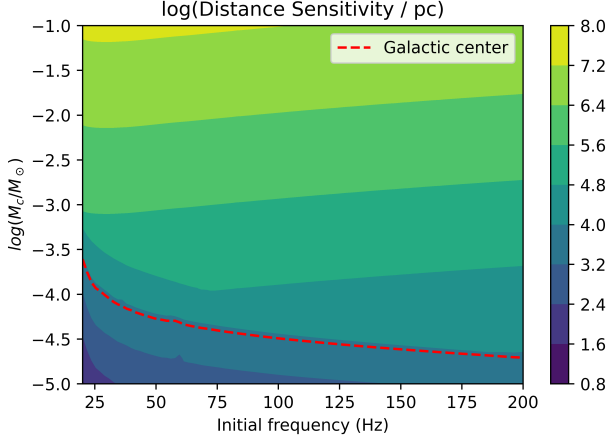


Figure 1. Maximum distance sensitivity of the resampling method as a function of the signal initial frequency f_0 and chirp mass M_c . The red dotted line corresponds to the distance of the galactic centre. This shows the vast majority of the parameter space has a maximum distance sensitivity beyond the galactic centre so could feasibly detect any primordial black holes within the galactic centre.

and chirp masses between $M_c = [10^{-4}, 10^{-1}]M_\odot$, the corresponding range is $\beta \sim [10^{-10}, 10^{-1}]$.

Because of the finite number of β templates, a signal with β_{signal} will be some small distance away from the closest template $\beta_{\text{template}} = \beta_{\text{signal}} + \Delta\beta$. When the signal is resampled with this closest template with some mismatch $\Delta\beta$, the signal will not be purely monochromatic but instead contain some dephasing. The spacing between β templates is chosen such that the dephasing is sufficiently small that the signal remains confined within a single frequency bin of the FFT. The dephasing of the signal is given by:

To find the optimal template spacing for β we calculate how much dephasing will be introduced by resampling a signal with β_{signal} with its closest template a small distance away $\beta_{\text{template}} = \beta_{\text{signal}} + \Delta\beta$. The mismatch $\Delta\beta$ means that the resampled signal will not be purely monochromatic but instead have a dephasing:

$$\begin{aligned} \Delta\phi &= \phi(\beta_{\text{signal}}, t) - \phi(\beta_{\text{template}}, t) \\ &= -\frac{6\pi}{5}f_0 \left(\frac{(1 - \frac{8}{3}\beta_{\text{signal}}t)^{5/8}}{\beta_{\text{signal}}} - \frac{(1 - \frac{8}{3}(\beta_{\text{signal}} + \Delta\beta)t)^{5/8}}{(\beta_{\text{signal}} + \Delta\beta)} \right) \end{aligned} \quad (24)$$

$$(25)$$

where we have substituted the phase evolution given in Eq. 3. The spacing of the template grid in β is chosen such that the dephasing after resampling is no more than some threshold $\Delta\phi_{\text{thr}}$, commonly taken to be $\pi/4$ as described in Ref. Find ref, so that the signal power is still mostly confined in a single FFT bin after the resampling. To analytically calculate the condition on $\Delta\beta$ we Taylor expand around small values of $\Delta\beta$ to get: Check the ϕ_0

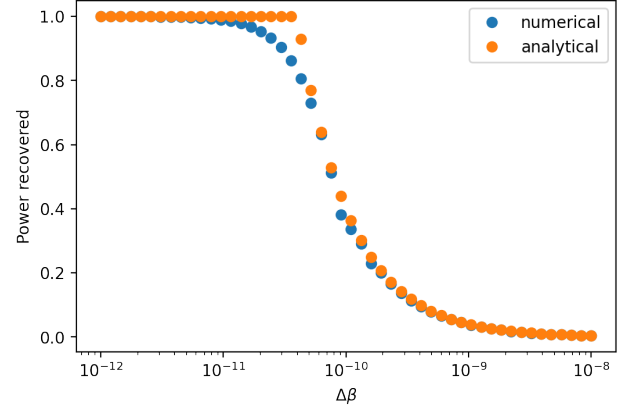


Figure 2. Fraction of power recovered from analytical predictions compared to numerically performing resampling. For a system with parameters: $f_0 = 20\text{Hz}$, $M_c = 10^{-3}M_\odot$, $T_{\text{obs}} = 0.5$ days.

term

$$\Delta\phi(t) \approx \frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta^2} \frac{-1 + \beta t}{(1 - \frac{8}{3}\beta t)^{3/8}} + \frac{6\pi f_0}{5} \frac{\Delta\beta}{\beta^2} \quad (26)$$

where the second term is to ensure there is no dephasing at $t = 0$. The dephasing is a monotonic function in time and the greatest dephasing is at the final time. Therefore, the time independent condition for $\Delta\beta$ is:

$$\Delta\beta \leq \frac{5\beta^2 \Delta\phi_{\text{thresh}}}{6\pi f_0} \left(\frac{(-1 + \beta t_{\text{max}})}{(1 - \frac{8}{3}\beta t_{\text{max}})^{3/8}} + 1 \right)^{-1}. \quad (27)$$

The 2nd order Taylor term is negligible over the range of β we are interested in searching over should this go in an Appendix?.

We can calculate the fraction of signal power lost as a result of the dephasing by calculating the time dependent frequency error:

$$\Delta f(t) = \frac{1}{2\pi} \frac{d[\Delta\phi(t)]}{dt} \quad (28)$$

$$= \frac{3f_0 \Delta\beta}{5\beta \left(1 - \frac{8\beta t}{3}\right)^{3/8}} \left(1 + \frac{-1 + \beta t}{1 - \frac{8\beta t}{3}} \right). \quad (29)$$

For a given β and signal duration, we define $t_{\text{crossover}}$ as the time when the frequency error will exceed the width of a FFT bin ($1/T$) after which signal power will be lost to neighbouring FFT bins. We can numerically solve for $t_{\text{crossover}}$ and compute the power of the signal after $t_{\text{crossover}}$:

$$\text{Power loss} = (h(t_{\text{crossover}} < t < T) | h(t_{\text{crossover}} < t < T)) \quad (30)$$

$$= \frac{16}{d^2} \left(\frac{5}{96} \right)^2 \frac{\beta^2 c^2}{\pi^4 f_0^4} \int_{t_{\text{crossover}}}^T \left(1 - \frac{8}{3}\beta t \right)^{-1/2} dt \quad (31)$$

or the fraction of total signal power lost is:

$$L = \text{Power loss fraction} = \frac{\int_{t_{\text{crossover}}}^T (1 - \frac{8}{3}\beta t)^{-1/2} dt}{\int_0^T (1 - \frac{8}{3}\beta t)^{-1/2} dt}, \quad (32)$$

$$= \frac{(1 - \frac{8}{3}\beta t_{\text{crossover}})^{1/2} - (1 - \frac{8}{3}\beta T)^{1/2}}{1 - (1 - \frac{8}{3}\beta T)^{1/2}} \quad (33)$$

We actually allow the frequency error to be around $(2/T)$ since the spectrum is still sharply peaked even if the dephasing has moved the mean by one FFT bin. We can compare this analytical calculation to real numerical implementations of resampling; where we simulate a signal with fixed parameters, perform resampling with a known mismatch $\Delta\beta$, perform the FFT and check how much of the total power has been recovered. Figure 2 compares the analytical and numerical calculations for a system with $f_0 = 20\text{Hz}$, $M_c = 10^{-3}M_\odot$, $\beta = 1.7 \times 10^{-8}$, $T_{\text{obs}} = 0.5$ days for a range of $\Delta\beta$. It shows a good level of agreement and verifies the analytical calculation.

If we want to fix the power loss due to template mismatches and compute the spacing between templates, we can invert Eqs. X-Y to get:

$$t_{\text{cross}} = \frac{3}{8\beta} \left(1 - \left(L + (1-L)(1 - \frac{8}{3}\beta T)^{1/2} \right)^2 \right) \quad (34)$$

$$\Delta\beta = \frac{2}{T} \frac{5\beta(1 - \frac{8}{3}\beta t_{\text{cross}})^{3/8}}{3f_0} \left(\frac{-1 + \beta t_{\text{cross}}}{1 - \frac{8}{3}\beta t_{\text{cross}}} + 1 \right)^{-1} \quad (35)$$

B. Grid in time

Both the duration of the signal (T) and the time at which the signal first enters the detector sensitivity band t_0 are not known a-priori. The duration of the signal can be calculated using Eq. 4 where the upper frequency limit is taken to be $f_{\text{max}} = 200\text{Hz}$. This upper limit is lower than the conventional 2kHz maximum frequency of the LVK detector sensitivity because the long-lived nature of these signals means that the greater sampling frequency needed to the high frequency components of the signal puts much greater computational burdens on the data analysis. In particular, the extra power emitted between 200-2000Hz can be computed analytically and is between

We first compute the power loss that can result from the unknown signal duration. Signals with $f_0 = [20, 199]\text{Hz}$ and chirp masses between $M_c = [10^{-5}, 10^{-1}]M_\odot$ have durations that span from $[10^2, 10^{12}]s$ although recall that the entire duration of the O4 observing run is expected to span 20 months ($\sim 5 \times 10^7 s$) with a duty cycle much lower than 100% due to one planned period of commissioning and detectors being out of observing mode. To search for these variable duration signals we fix a window width W .

Signals with durations shorter than this window $T < W$ will have their signal to noise ratio suppressed by a factor of W/T . If the signal duration is longer than the window ($T > W$), we can semicoherently combine the analysis from different windows which suppressed the signal-to-noise ratio by a factor of $\sqrt{T/W}$ [30]. The effect of this decrease in SNR on the distance sensitivity is:

$$d_{\text{measured}} = \begin{cases} \sqrt{\frac{T}{W}} d_{\text{opt}} & T < W \\ \sqrt[4]{\frac{W}{T}} d_{\text{opt}} & W < T \end{cases}. \quad (36)$$

Since this pipeline is a targeted search, the absolute distance sensitivity is not as important as the amount of the parameter space which can reach a particular target of interest. This is because sensitivity at distances beyond the target do not cover regions of space with much dark matter density and so provide little marginal probability of detecting primordial black holes. We therefore choose a window width W which can preserve the maximum fraction of parameter space capable of reaching the galactic center. I.e. we use Eq. 36 to produce distance sensitivity plots similar to Fig 1 for different values of W . For each W we compute the area of the parameter space which is still sensitive to the galactic center and numerically solve for the W which maximises this area. Needs to be integer number of sidereal days to ensure 5-vector statistic works

VII. CONCLUSION

Here's what you should learn now that I done'd it.

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